

Practice Exam #1

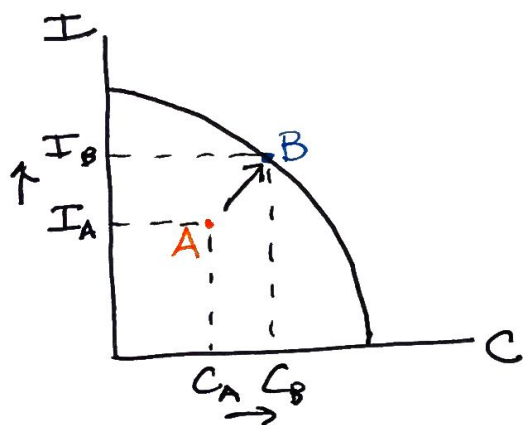
SUGGESTED SOLUTIONS

- ① 2 goods — consumer goods, investment goods

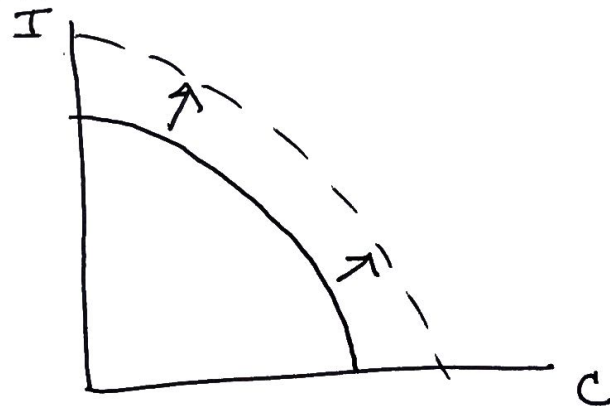


Uncertain

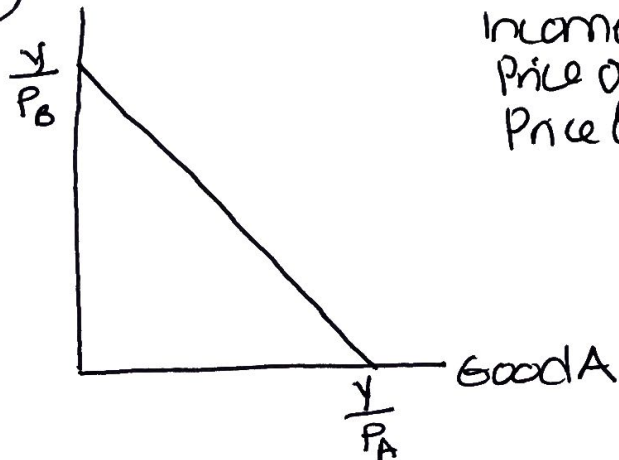
There are two ways an increase in both C & I could occur. First, the economy could start at A due to unemployment of resources & move to B as it uses those existing resources



Alternatively, the policy could have increased resources (allowed for greater immigration thereby increasing labor, for example). This would shift the entire PPF outwards.



- ② Good B



Income = y

Price of A = P_A

Price of B = P_B

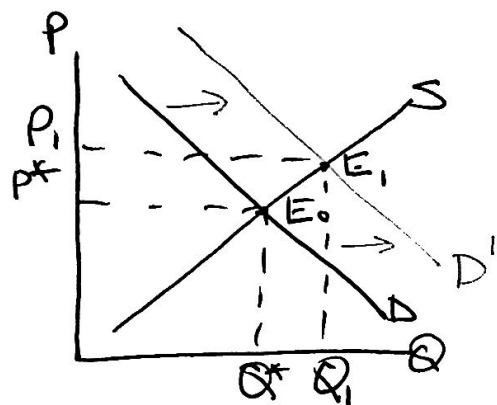
True

To see this:

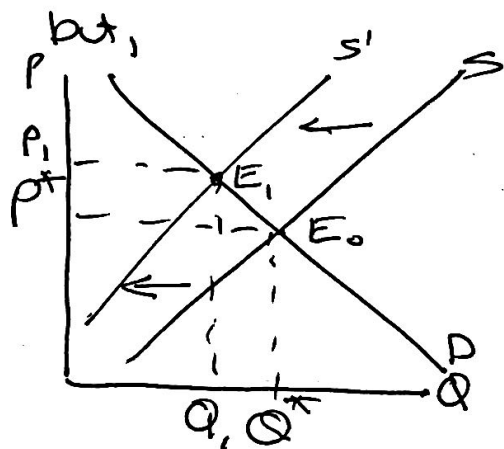
$$\text{slope of BC} = \frac{y/P_B}{y/P_A} = \frac{y/P_B}{y/P_A} = \frac{P_A}{P_B}$$

$$\text{slope of BC} = \frac{P_A}{P_B}$$

- ③ The price in the market rose.
Let's examine the ways that could happen.



⇒ D↑ does lead to P* ↑ to P₁



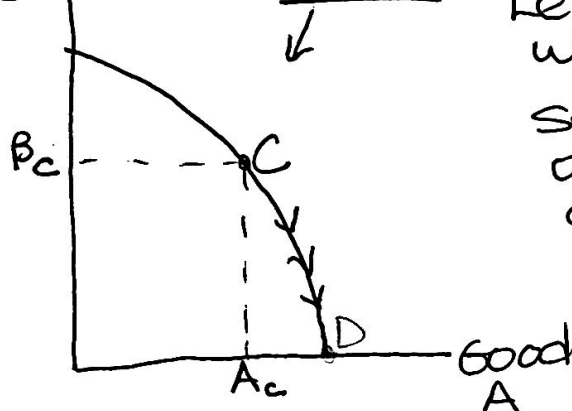
⇒ S↓ also leads to P* ↑ to P₁

⇒ Uncertain

④ True

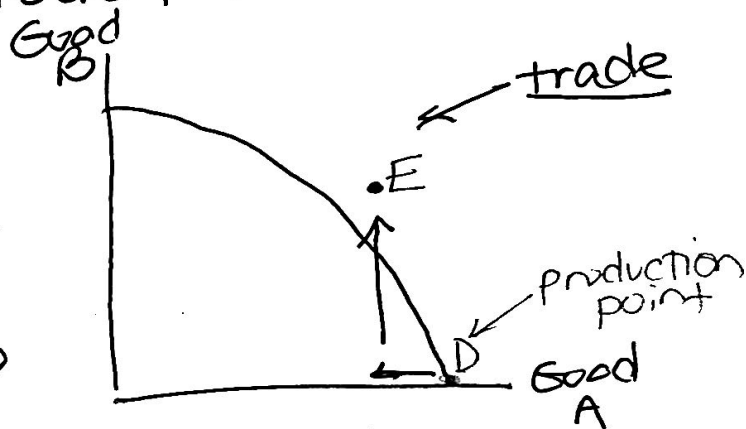
Rational decisions are made by comparing MC & MB. If going to a discount store has higher costs due to crowding & being understaffed than it may prevent them from shopping there (MC > MB) whereas at a convenience store, the MC may be lower, thereby inducing them to shop (MB > MC). Prices may not matter (impact decision making) as much for high income individuals.

⑤ Good B no trade



Let's consider how specialization will impact the PPF to the left. Specialization means you produce one good (or at least more of one good). If the individual originally produced at pt C, then complete specialization in good A would move the production point to D.

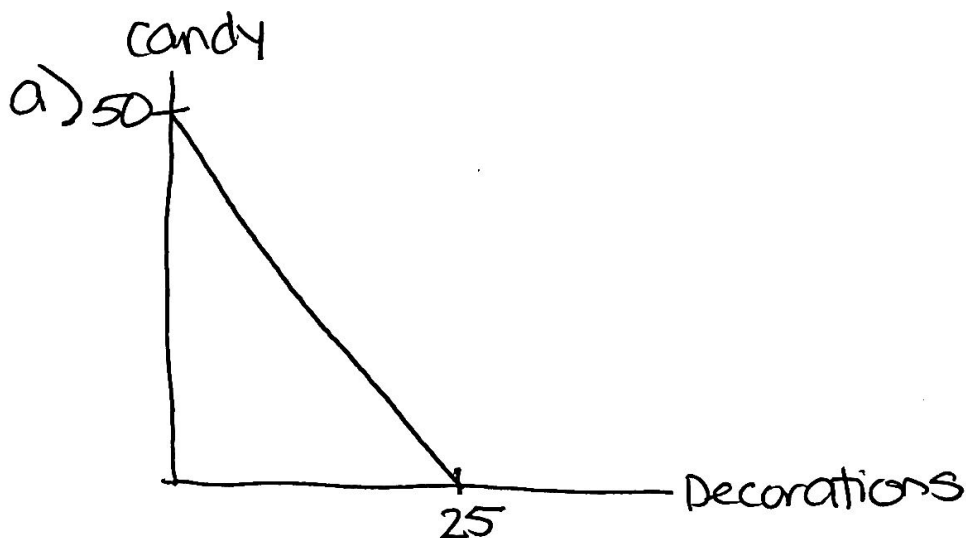
If this individual were to trade, they would produce whichever good they have a comparative advantage in (let's say Good A). They would produce at point D but could now sell some of Good A in exchange for Good B & consume at some other point, E. While they can now consume at a point outside the PPF, it doesn't shift it because it had no impact on their production capabilities (i.e. resources or technology).



False

Problems

① $Y = \$50$
 $P_c = \$1$
 $P_d = \$2$



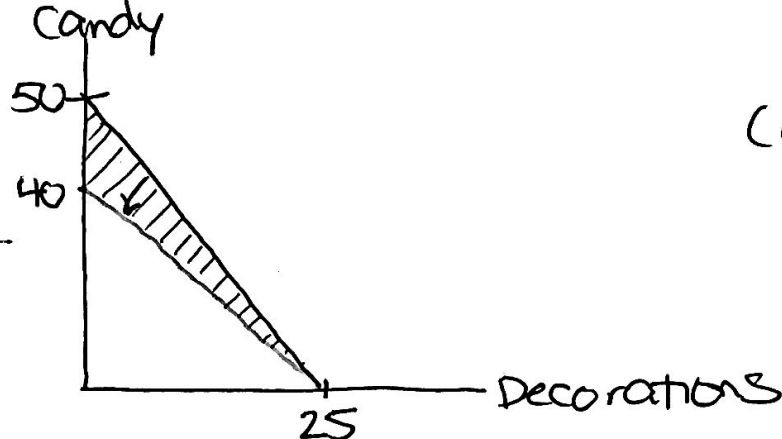
b) opp cost_{candy} = $\frac{25 \text{ decor}}{50 \text{ candy}} = \frac{1}{2} \text{ decor}$

$OC_c = \frac{1}{2} \text{ decor}$

opp cost_{decor} = $\frac{50 \text{ candy}}{25 \text{ decor}} = 2 \text{ candy}$

$OC_d = 2 \text{ candy}$

c) $P_c = \$1.25$
 $P_D = \$2$
 — = new budget constraint



(could draw this on same B.C. as in (b)).

The opp costs do change since the slope of the budget constraint changed.

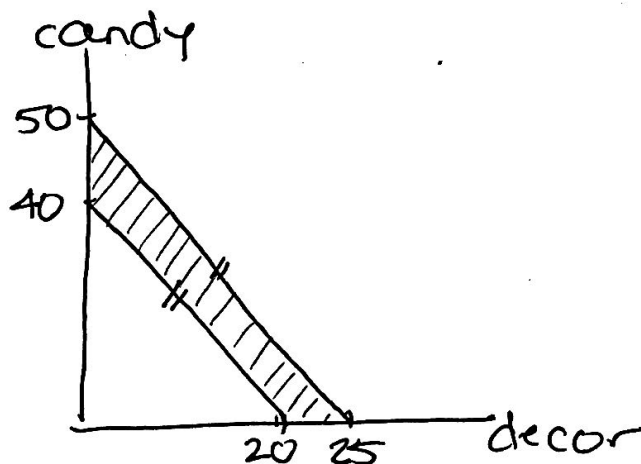
$$\text{opp cost}_{\text{decor}} = \frac{40 \text{ candy}}{25 \text{ decor}} = 1.6 \text{ candy}$$

$OC_d = 1.6 \text{ candy}$
$OC_c = .625 \text{ decor}$

$$\text{opp cost}_c = \frac{1}{\text{opp cost}_{\text{decor}}} = .625 \text{ decor}$$

Your consumption possibilities also changed (they are reduced) since candy is now more expensive. Consumption bundles are reduced by the shaded area.

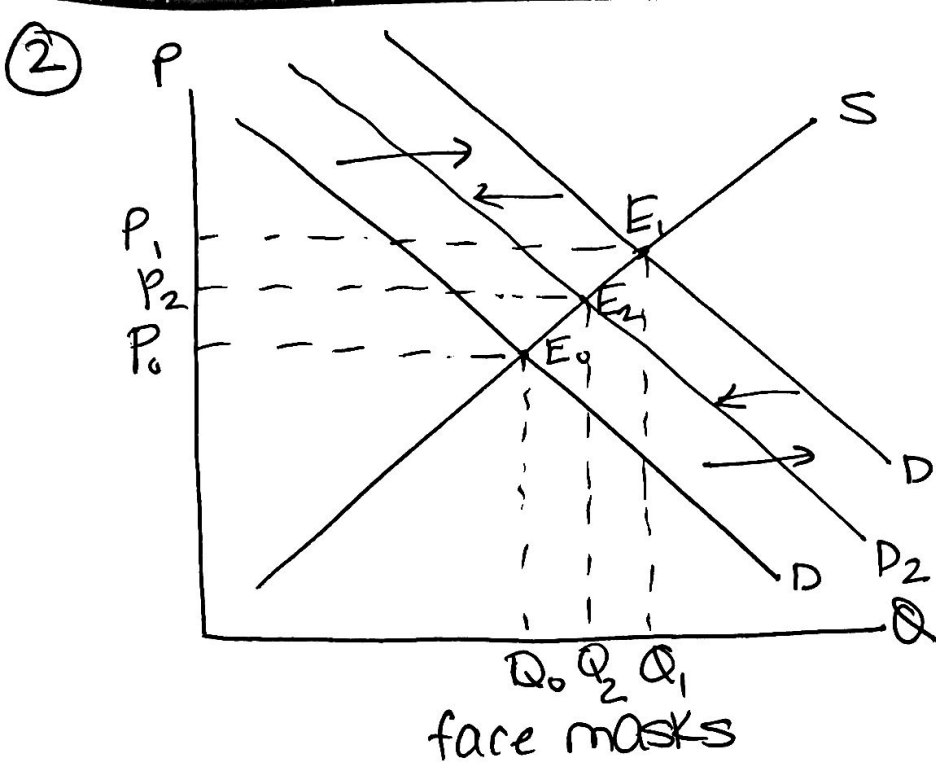
d) $P_c = \$1$
 $P_D = \$2$
 $Y = \$40$
 — = new budget constraint



the slope doesn't change so opp costs are unchanged. can also see this by calculation:

$$OC_d = \frac{40}{20} = 2 \text{ candy} \Rightarrow OC_c = \frac{1}{2} \text{ decor}$$

The new possible consumption bundles are reduced however due to the lower income. Fewer goods can be purchased. Consumption bundles are reduced by the shaded area.



- ① masks are encouraged / mandated.
 - consumer tastes / preferences
 - $D \uparrow$ to D_1
- ② people lost jobs / more uncertain about jobs
 - income of buyers & expected income of buyers
 - $D \downarrow$ to D_2

Since $D \uparrow$ & then $\downarrow$ both P & Q are indeterminate; it depends on the relative size of the two shifts. Given the mandating of masks & the health benefits (or protection) during a pandemic, I believe the $\uparrow$ in D is likely to be larger than the $\downarrow$ in D due to income.

Transitions:

As $D \uparrow$ due to tastes, at P_0 , $Q_d > Q_s$ leading to a shortage. Consumers compete for masks, driving the price up to P_1 & the new eq. point of E_1 .

At E_1 , $D \downarrow$ due to lower income & / or exp future incomes.

At P_1 , $Q_d < Q_s$ now creating a surplus. This puts downward pressure on price as sellers compete to sell their masks, moving price to P_2 (at eq. pt E_2)

③

	bread	pie	
monica	2	1	
Chandler	4	4	← labor hrs

↓ 10 labor hrs

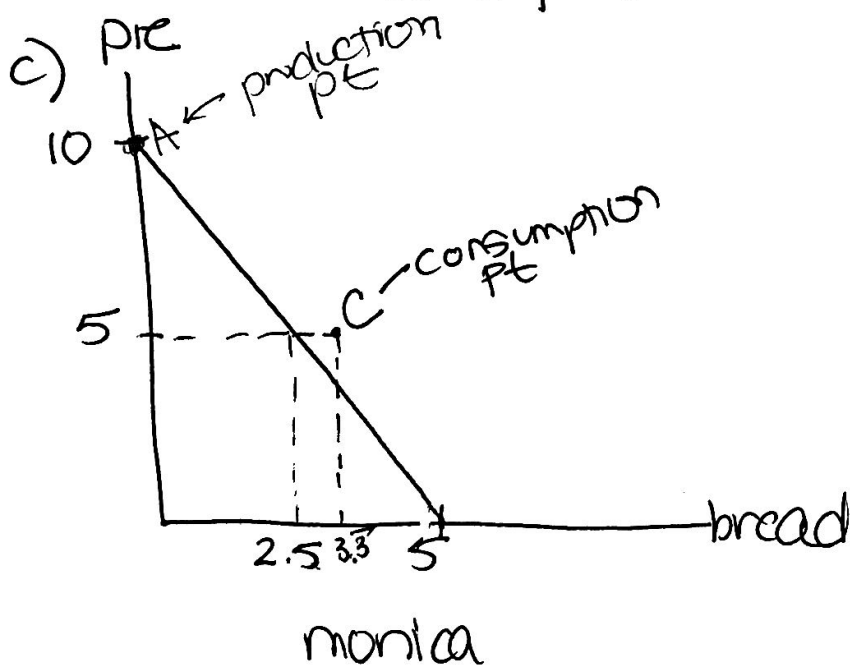
	5	10	
monica			
Chandler	2.5	2.5	← output

a) $OC_{\text{bread}}^C = \frac{2.5 \text{ pie}}{2.5 \text{ bread}} = 1 \text{ pie}$ $OC_B^C = 1 \text{ pie}$

$OC_{\text{bread}}^M = \frac{10 \text{ pie}}{5 \text{ bread}} = 2 \text{ pie}$ $OC_B^M = 2 \text{ pies}$

b) If they specialize in order to maximize combined output, they should each produce what they have a comparative advantage in.

⇒ monica should produce pies; Chandler should produce bread.



d) Trade can occur at a price that is between the two opp costs. Let's set the trade price of bread at 1.5 pies

$\text{trade price}_B = 1.5 \text{ pies}$

monica can sell 5 pies & get 3.3 bread and consume at C

④ Economic decisions are made at the margin. With each unit of pollution we decide to eliminate, we should compare the costs & benefits.

At high levels of pollution the MB of reducing pollution are likely to be relatively high, & the MC relatively low.

As long as $MB > MC$, we should reduce pollution.

However, the less pollution there is, the harder it is likely to be to eliminate (MC get higher).

Meanwhile the ^{marginal} benefits of less pollution are

likely smaller. Thus, eliminating all pollution is unlikely to be a rational economic decision.